

**Topic: Light as electromagnetic wave, The Poynting Vector, Interference of light, Young's double slit experiment**

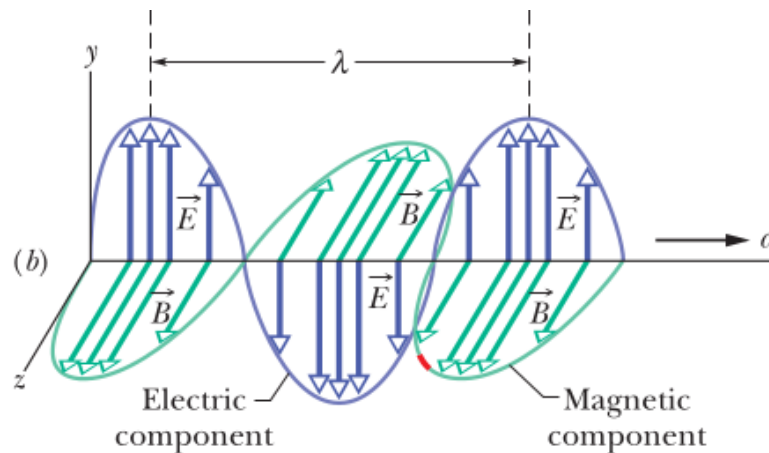
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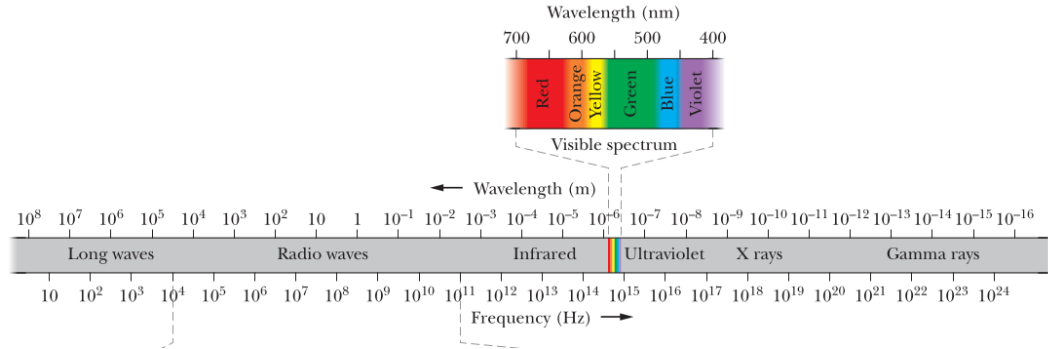
## Electromagnetic Nature of Light

Light is an electromagnetic wave, as explained by James Clerk Maxwell's equations. The key features of light as an electromagnetic wave are:

- **Electric and Magnetic Fields:** Light consists of perpendicular oscillating electric ( $E$ ) and magnetic ( $B$ ) fields.
- **Transverse Wave:** Both fields are transverse to the direction of wave propagation.
- **Mutual Regeneration:** A changing electric field generates a magnetic field, and a changing magnetic field generates an electric field.

**YOUTUBE LINK:** <https://youtu.be/Oo8XMxXY32g?si=2BcGRmIlGMgGWth0>





## The Poynting Vector

The Poynting vector, named after the British physicist John Henry Poynting, is a vector that represents the **energy transfer per unit time and per unit area** or power flow of an electromagnetic field. It is crucial in understanding how electromagnetic waves transfer energy through space.

## Definition of the Poynting Vector

The Poynting vector  $\vec{S}$  is defined as:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

where:

- $\vec{E}$  is the electric field vector (in volts per meter, V/m),
- $\vec{B}$  is the magnetic field vector (in teslas, T),
- $\mu_0$  is the permeability of free space, with a value of  $4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ .

## Interference of Light

Interference is a phenomenon in which two or more waves superpose to form a resultant wave of greater, lower, or the same amplitude. In the case of light waves, interference occurs when two coherent light waves overlap and combine.

The principle of superposition states that when two or more waves meet at a point, the resultant displacement is the algebraic sum of the individual displacements.

### Conditions for Interference

For sustained interference to occur, the following conditions must be satisfied:

1. The two light sources must be coherent (i.e., they must have a constant phase difference).
2. The light waves must have nearly the same frequency and wavelength.
3. The superimposing light waves must have similar amplitudes.
4. The light sources must be monochromatic (single wavelength).

### Classification of Interference

When two coherent waves interfere, the resultant intensity at a point depends on their phase difference:

- **Constructive Interference:** When two waves meet in phase, their amplitudes add up, leading to an increase in intensity. Mathematically, the path difference must be:

$$\Delta x = m\lambda, \quad m = 0, 1, 2, 3, \dots$$

- **Destructive Interference:** When two waves meet out of phase, their amplitudes cancel out, leading to a decrease in intensity. The path difference must be:

$$\Delta x = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, 1, 2, 3, \dots$$

## Young's Double Slit Experiment

Young's double-slit experiment, first performed by Thomas Young in 1801, is a demonstration of the wave nature of light. The basic arrangement of the experiment involves a monochromatic light source, slits, and interference patterns that emerge due to the superposition of light waves.

### Experimental Setup

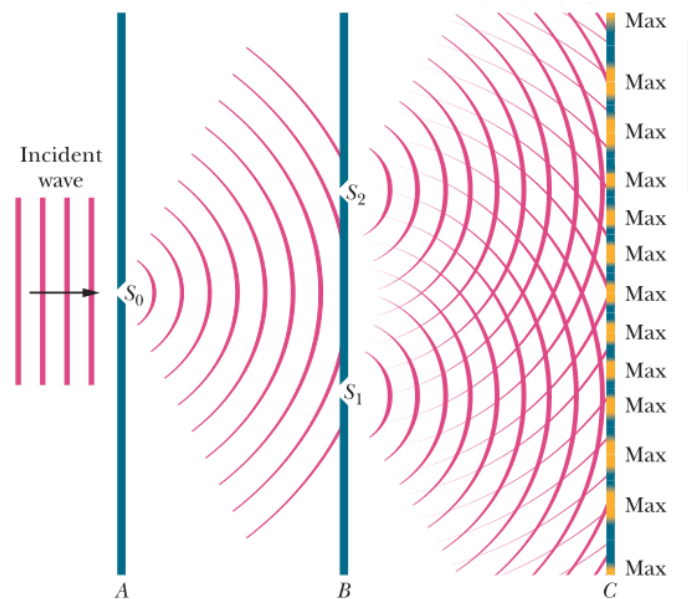
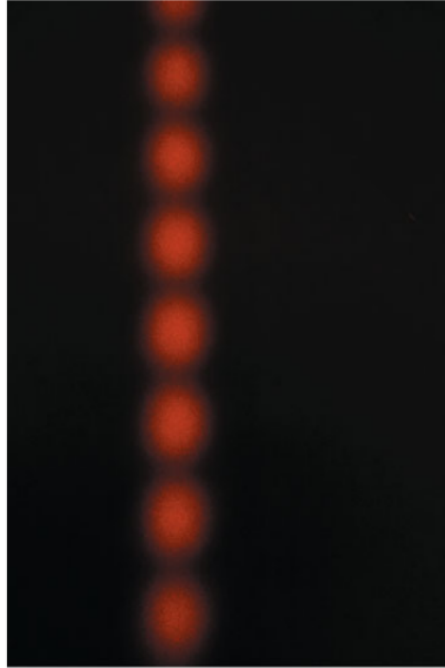


Figure 1: The waves emerging from the two slits overlap and form an interference pattern.

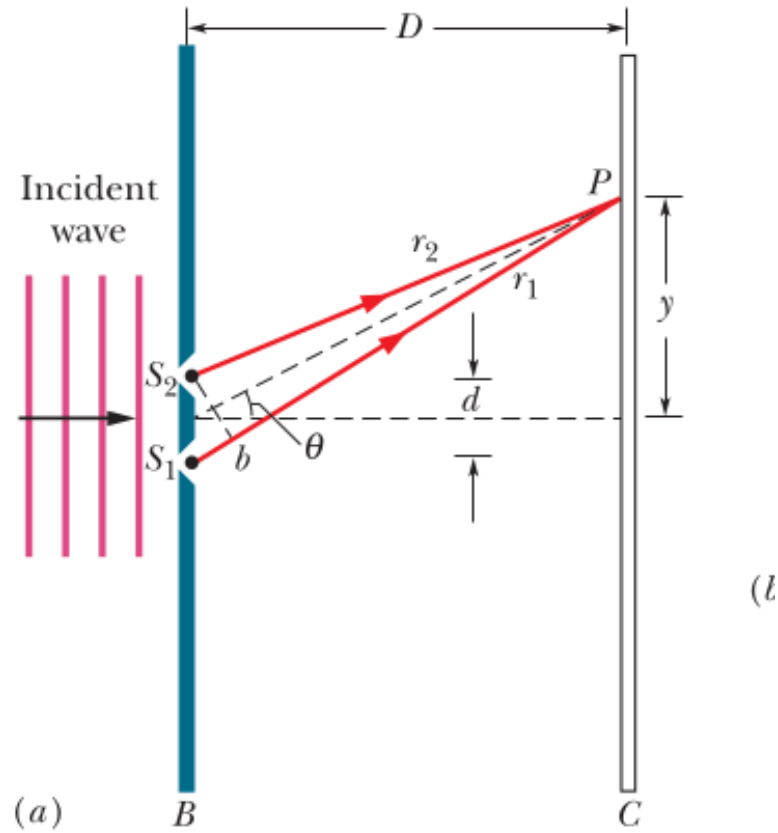
The basic arrangement of Young's experiment is as follows:

- Light from a distant monochromatic source illuminates slit  $S_0$  on screen A.
- The light emerging from slit  $S_0$  spreads via diffraction and illuminates the two slits  $S_1$  and  $S_2$  on screen B.
- The slits  $S_1$  and  $S_2$  cause the light to diffract, generating circular waves that overlap beyond screen B.
- These overlapping waves interfere with each other, creating an interference pattern of alternating bright and dark fringes on a screen placed beyond screen B.

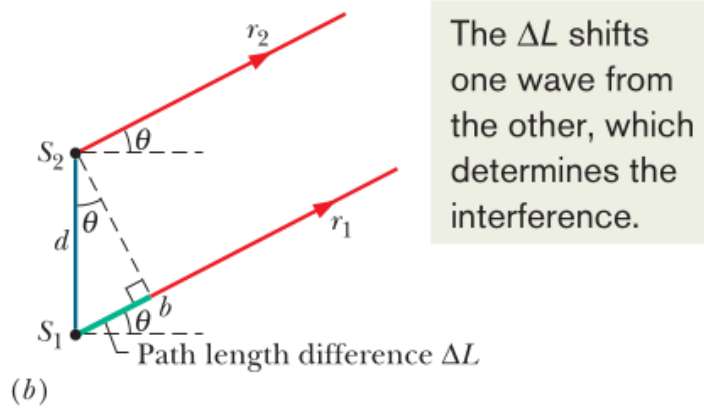
The interference occurs due to the superposition of the light waves coming from slits  $S_1$  and  $S_2$ . Constructive interference leads to bright fringes, while destructive interference results in dark fringes.



## Path Length Difference and Fringe Conditions



In Young's double-slit experiment, the path difference between the light from the two slits results in interference patterns. For simplicity, we assume that the distance from the slits to the screen is much greater than the slit separation,  $D \gg d$ . Under this assumption, we can approximate the rays emerging from the slits as being parallel to each other and at an angle  $\theta$  to the central axis.



The path length difference  $\Delta L$  is related to the angle  $\theta$  by the equation:

$$\Rightarrow \Delta L = d \sin \theta$$

## Condition for Bright Fringes (Maxima)

For a bright fringe, the path length difference must be either zero or an integer multiple of the wavelength. We can write this requirement as:

$$\Delta L = d \sin \theta = m\lambda$$

where  $m = 0, 1, 2, \dots$  is the integer representing the order of the maxima (bright fringes). Thus, the condition for bright fringes becomes:

$$d \sin \theta = m\lambda \quad (\text{for maxima, bright fringes})$$

where  $\lambda$  is the wavelength of the light.

## Condition for Dark Fringes (Minima)

For a dark fringe, the path length difference must be an odd multiple of half a wavelength. , we can write this requirement as:

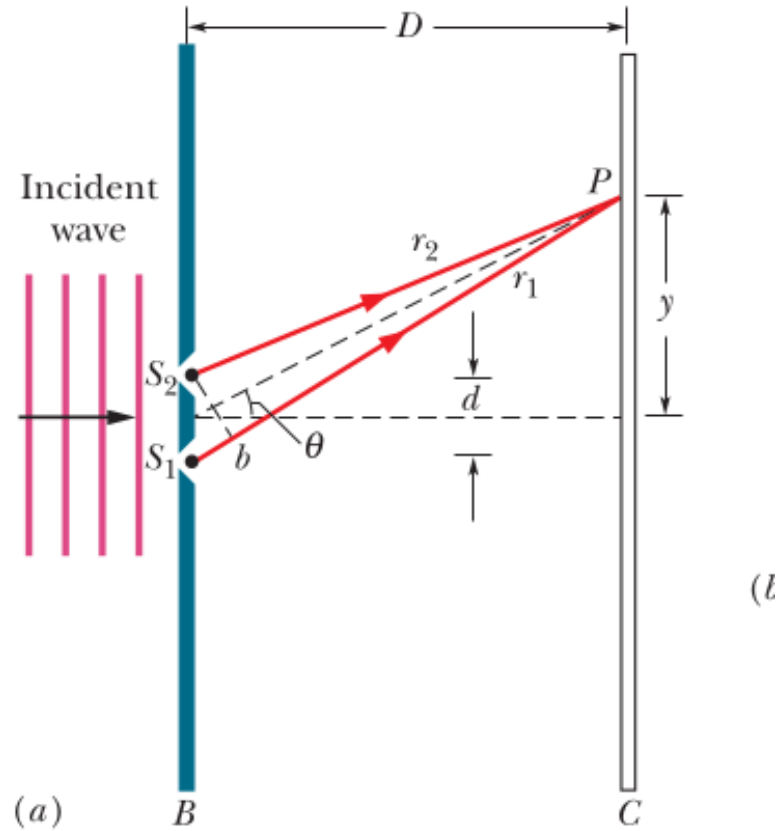
$$\Delta L = d \sin \theta = \left(m + \frac{1}{2}\right) \lambda$$

where  $m = 0, 1, 2, \dots$  is the integer representing the order of the minima (dark fringes). Thus, the condition for dark fringes becomes:

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda \quad (\text{for minima, dark fringes})$$

### Problem : 1

Find the distance on screen C between adjacent maxima near the center of the interference pattern. The wavelength of the light is  $\lambda = 546 \text{ nm}$ , the slit separation is  $d = 0.12 \text{ mm}$ , and the slit-screen separation is  $D = 55 \text{ cm}$ . Assume that the angle  $\theta$  in the figure is small enough to permit the use of the approximations  $\sin \theta \approx \tan \theta \approx \theta$ , where  $\theta$  is expressed in radian measure.



### Solution

To find the distance between adjacent maxima, we first express the angle  $\theta$  for the  $m$ -th maximum in terms of the other given quantities. For a bright fringe, the path length difference must be an integer multiple of the wavelength. From the equation for the  $m$ -th maximum in Young's double-slit experiment, we have:

$$d \sin \theta = m\lambda$$



Using the small angle approximation  $\sin \theta \approx \theta$ , we can write:

$$d\theta = m\lambda \quad \Rightarrow \quad \theta = \frac{m\lambda}{d}$$

Next, we calculate the vertical distance  $y_m$  of the  $m$ -th maximum from the center of the pattern using the geometry of the setup. The vertical distance is related to the angle  $\theta$  by:

$$y_m = D \tan \theta$$

Since  $\tan \theta \approx \theta$  for small angles, we have:

$$y_m = D\theta = D\frac{m\lambda}{d}$$

For the next maximum, where  $m$  increases by 1, the vertical distance  $y_{m+1}$  is:

$$y_{m+1} = D\frac{(m+1)\lambda}{d}$$

The distance between adjacent maxima is:

$$\Delta y = y_{m+1} - y_m = D\left(\frac{(m+1)\lambda}{d} - \frac{m\lambda}{d}\right)$$

$$\Delta y = D\frac{\lambda}{d}$$

Substituting the given values:

$$\lambda = 546 \times 10^{-9} \text{ m}, \quad D = 55 \times 10^{-2} \text{ m}, \quad d = 0.12 \times 10^{-3} \text{ m}$$

$$\Delta y = (55 \times 10^{-2} \text{ m}) \frac{546 \times 10^{-9} \text{ m}}{0.12 \times 10^{-3} \text{ m}}$$

$$\Delta y = 2.50 \times 10^{-3} \text{ m} = 2.5 \text{ mm}$$

## Problem : 2

In a double-slit experiment, the distance between slits is 5.0 mm and the slits are 1.0 m from the screen. Two interference patterns can be seen on the screen: one due to light of wavelength 480 nm, and the other due to light of wavelength 600 nm. What is the separation on the screen between the third-order ( $m = 3$ ) bright fringes of the two interference patterns?

### Solution:

The position of the bright fringes in a double-slit interference pattern is given by the equation:

$$y_m = \frac{m\lambda L}{d}$$

where: -  $y_m$  is the position of the  $m$ -th order bright fringe, -  $m$  is the order of the fringe (for the third-order,  $m = 3$ ), -  $\lambda$  is the wavelength of the light, -  $L$  is the distance from the slits to the screen, -  $d$  is the distance between the slits.

**Given:**  $d = 5.0 \text{ mm} = 5.0 \times 10^{-3} \text{ m}$ ,  $L = 1.0 \text{ m}$ ,  $\lambda_1 = 480 \text{ nm} = 480 \times 10^{-9} \text{ m}$ ,  $\lambda_2 = 600 \text{ nm} = 600 \times 10^{-9} \text{ m}$ ,  $m = 3$ .

For the third-order bright fringe, the position for each wavelength is:

$$y_{3,1} = \frac{3 \times 480 \times 10^{-9} \times 1.0}{5.0 \times 10^{-3}} = 0.288 \text{ mm}$$

$$y_{3,2} = \frac{3 \times 600 \times 10^{-9} \times 1.0}{5.0 \times 10^{-3}} = 0.360 \text{ mm}$$

The separation between the third-order fringes for the two wavelengths is:

$$\Delta y = y_{3,2} - y_{3,1} = 0.360 \text{ mm} - 0.288 \text{ mm} = 0.072 \text{ mm}$$

### Answer:

The separation between the third-order bright fringes of the two interference patterns is 0.072 mm.

### Problem : 3

In a double-slit arrangement, the slits are separated by a distance equal to 100 times the wavelength of the light passing through the slits.

[(a)] What is the angular separation in radians between the central maximum and an adjacent maximum? [(b)] What is the distance between these maxima on a screen 50.0 cm from the slits?

### Solution:

In a double-slit diffraction pattern, the angular position of the maxima is given by the condition:

$$d \sin \theta = m\lambda$$

where: -  $d$  is the distance between the slits, -  $\theta$  is the angle of diffraction, -  $m$  is the order of the maximum (for adjacent maxima,  $m = 1$  for the first maximum), -  $\lambda$  is the wavelength of the light.

**Given:** - The distance between the slits is  $d = 100\lambda$ , - The distance from the slits to the screen is  $D = 50.0 \text{ cm} = 0.50 \text{ m}$ .

#### Part (a): Angular Separation

For adjacent maxima, the angle  $\theta$  for the first order maximum ( $m = 1$ ) is:

$$d \sin \theta = \lambda$$

Substitute  $d = 100\lambda$ :

$$100\lambda \sin \theta = \lambda$$

$$\sin \theta = \frac{1}{100}$$

For small angles,  $\sin \theta \approx \theta$  (in radians), so:

$$\theta \approx \frac{1}{100} \text{ radians}$$

#### Part (b): Distance Between the Maxima on the Screen

The distance between two adjacent maxima on the screen is given by:

$$y = L \tan \theta$$

For small angles,  $\tan \theta \approx \sin \theta$ , so:

$$y \approx L \sin \theta = 0.50 \times \frac{1}{100} = 0.005 \text{ m} = 5.00 \text{ mm}$$